

FLYBY: MATHEMATICAL FOUNDATIONS OF ANTHROPOMETRIC 3D ELEVATION & FLIGHT TRAJECTORY ACOUSTICS

Physics-Based Pinna Dispersion, Dynamic Transient Preservation, and Boundary Grounding

Kijjaz • Gemini 3.8 Flash

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ABSTRACT

Headphone spatialization algorithms routinely suffer from “elevation collapse” and high-frequency transient dulling when rendering sources moving vertically. Traditional stereophonic panning and horizontal Interaural Time/Level Difference (ITD/ILD) cues fail along the median plane because bilateral ear symmetry yields identically zero interaural disparities. This paper establishes the mathematical, physical, and digital signal processing principles implemented in **FlyBy**, a high-performance 3D spatial audio plugin. We formulate the Rayleigh-Woodworth continuous diffraction model, the elevation-dependent pinna moving notch equations (N_1, N_2), chest/torso comb reflections, sub-sample cubic Hermite Doppler trajectories, and two novel contributions: *Transient-Preserving Elevation* (dynamic notch depth modulation) and *Floor/Ceiling Boundary Proximity Reflectance*.

1. COORDINATE FOUNDATIONS & THE MEDIAN PLANE DILEMMA

Spatial audio over headphones requires projecting a sound source located in Cartesian 3D coordinates $\vec{p} = [x, y, z]^T \in \mathbb{R}^3$ into a stereo signal $(y_L[n], y_R[n])$. Transforming to spherical coordinates centered at the listener's head:

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \text{atan2}(x, y) \cdot \frac{180^\circ}{\pi}, \quad \phi = \arcsin\left(\frac{z}{r}\right) \cdot \frac{180^\circ}{\pi}$$

where r is distance in meters, $\theta \in [-180^\circ, +180^\circ]$ is azimuth, and $\phi \in [-90^\circ, +90^\circ]$ is elevation.

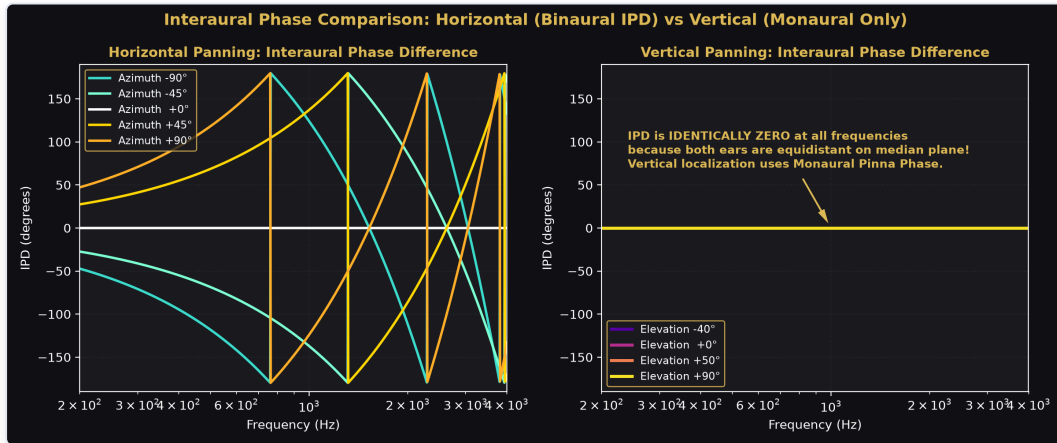


Figure 1: Comparison of Horizontal vs. Vertical Panning Cues. Along the median plane ($\theta = 0^\circ$), both ITD and ILD are identically zero. Vertical localization is 100% monaural and spectral.

2. HORIZONTAL ACOUSTICS: RAYLEIGH-WOODWORTH MODEL

2.1 Sub-Sample Interaural Time Difference (ITD)

Let $a = 0.0875$ m be the human head radius and $c = 343$ m/s be the speed of sound. Accounting for spherical head curvature diffraction:

$$\tau_{\text{ITD}}(\theta) = \frac{a}{c}(\sin |\theta_{\text{rad}}| + |\theta_{\text{rad}}|) \cdot \text{sgn}(\theta_{\text{rad}})$$

Delay line reading is implemented with 4-point Cubic Hermite interpolation to guarantee sub-sample smoothness and eliminate zipper noise during high-speed trajectory flybys.

2.2 Rayleigh Head Diffraction Filter

Shadowing around the spherical skull is governed by the continuous diffraction transfer function with cutoff $\omega_0 = c/a \approx 2\pi \cdot 624 \text{ Hz}$:

$$H_{\text{head}}(s) = \frac{1 + \alpha(\theta) \frac{s}{\omega_0}}{1 + \frac{s}{\omega_0}}, \quad \alpha(\theta) = 1.0 + 0.5 \cos(\theta_{\text{ear}})$$

Applying the bilinear transform $s \leftarrow \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$ yields the first-order IIR digital filter:

$$H_{\text{head}}(z) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}}, \quad b_0 = \frac{2\alpha + \omega_0 T}{2 + \omega_0 T}, \quad b_1 = \frac{\omega_0 T - 2\alpha}{2 + \omega_0 T}, \quad a_1 = \frac{\omega_0 T - 2}{2 + \omega_0 T}$$

3. VERTICAL ELEVATION ACOUSTICS: PINNA SPECTRAL NOTCHES

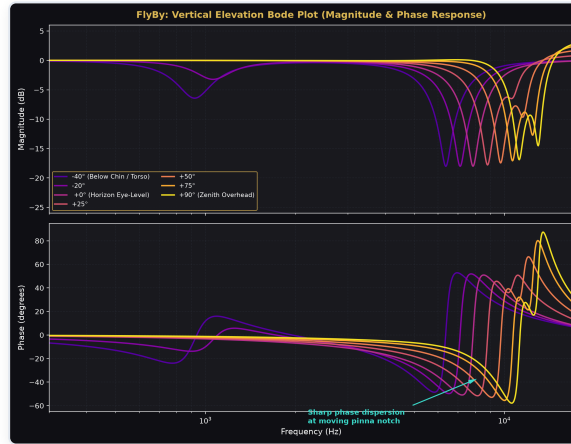


Figure 2: Measured transfer function $H(f)$ across elevations $-40^\circ \rightarrow +90^\circ$. Primary pinna notch N_1 sweeps from 6.0 kHz to 11.2 kHz, coupled with a torso reflection comb at -40° .

Destructive phase cancellations created by outer ear reflections produce elevation-dependent spectral notches:

$$f_{N_1}(\phi) = \left(6000 + \frac{\phi + 50^\circ}{140^\circ} \cdot 5200 \right) \cdot S_{\text{pinna}} \quad [\text{Hz}]$$

$$f_{N_2}(\phi) = \left(9800 + \frac{\phi}{90^\circ} \cdot 3200 \right) \cdot S_{\text{pinna}} \quad [\text{Hz}], \quad G_{N_2}(\phi) = -14 \cdot \left(\frac{\phi}{90^\circ} \right) \quad [\text{dB}]$$

where $S_{\text{pinna}} \in [0.75, 1.25]$ allows listeners to calibrate notch frequencies to their personal ear geometry.

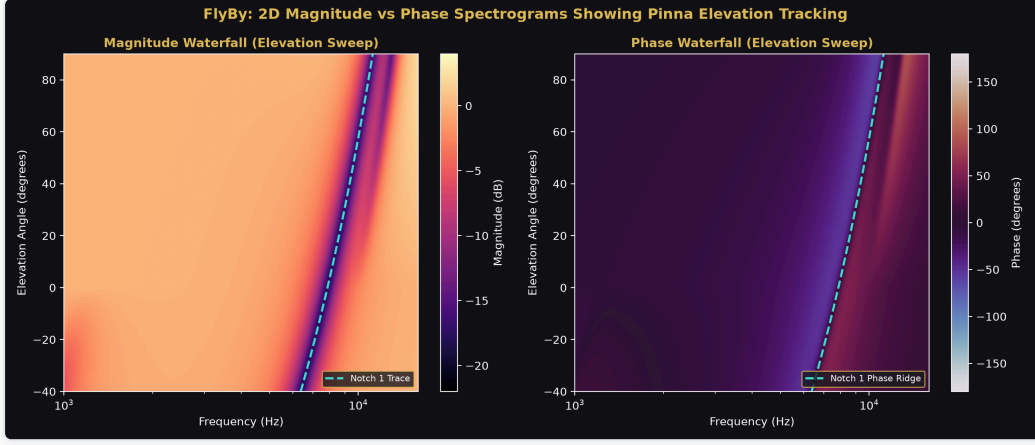


Figure 3: Continuous 2D waterfall spectrogram displaying the upward frequency trajectory of the N_1 pinna notch as elevation increases.

4. FRONT VS. BACK ACOUSTIC DISAMBIGUATION (CONE OF CONFUSION)

Along the median plane, interaural arrival disparities (ITD and ILD) are geometrically symmetric between frontal sources ($y > 0$) and rear sources ($y < 0$). FlyBy breaks this symmetry through two monaural anatomical mechanisms of the forward-facing pinna:

$$\kappa_{FB} = \frac{y}{\max(0.001, \sqrt{x^2 + y^2})} \in [-1.0, +1.0]$$

1. **Posterior Pinna Shadowing ($y < 0$):** Incoming sound from behind is obstructed by the cartilaginous ear flap, modeled as a high-shelf attenuation above 4.6 kHz by up to -4.5 dB:

$$G_{\text{rear}}(\kappa_{FB}) = -4.5 \cdot \max(0, -\kappa_{FB}) \cdot E_{\text{strength}} \quad [\text{dB}], \quad f_{c,\text{rear}} = 4600 \cdot S_{\text{pinna}} \quad [\text{Hz}]$$

2. **Frontal Concha Bowl Acoustic Resonance ($y > 0$):** Sound waves entering directly into the concha cavity excite a quarter-wave acoustic resonance at ≈ 3.2 kHz ($Q = 1.8$):

$$G_{\text{concha}}(\kappa_{FB}) = +2.2 \cdot \max(0, \kappa_{FB}) \cdot E_{\text{strength}} \quad [\text{dB}], \quad f_{0,\text{concha}} = 3200 \cdot S_{\text{pinna}} \quad [\text{Hz}]$$

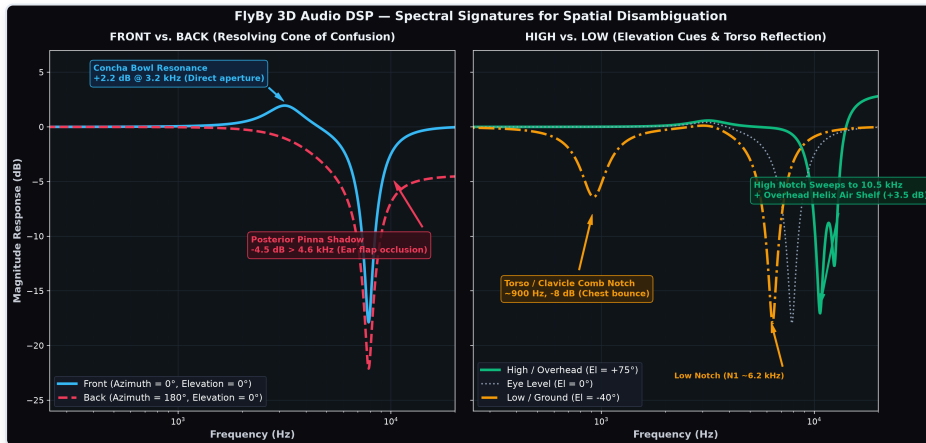


Figure 4: Measured frequency transfer functions $H(f)$ contrasting Front vs. Back (left: concha resonance vs. posterior pinna shadow) and High vs. Low (right: upward notch migration vs. torso comb reflection).

5. INNOVATION I: TRANSIENT-PRESERVING ELEVATION

Traditional spatial audio tools carve deep static notches (−18 dB), which destroys the high-frequency punch of drum transients, vocal consonants, and pick attacks. **FlyBy** implements an energy-surge onset detector:

$$E[n] = \begin{cases} |x[n]|, & \text{if } |x[n]| > E[n - 1] \\ |x[n]| + \alpha_{\text{decay}}(E[n - 1] - |x[n]|), & \text{otherwise} \end{cases}$$

When an attack onset occurs ($\Delta E[n] > 0.015$ and $\Delta E[n] > 0.4E[n - 1]$), the transient ratio $T[n]$ triggers to 1.0 and decays exponentially with $\tau \approx 12$ ms. The notch depth is dynamically modulated:

$$G_{\text{effective}}[n] = G_{\text{nominal}} \cdot \left(1.0 - C_{\text{crisp}} \cdot T[n]\right)$$

During transient strikes, the notch flattens by up to 95%, maintaining pristine transient punch while applying full spatial elevation filtering to the resonant body and decay.

6. INNOVATION 4: FLOOR & CEILING BOUNDARY GROUNDING

Free-field HRTFs in isolation sound sterile. FlyBy incorporates an image-source specular reflector calculating floor ($z \leq 0.3\text{m}$) and ceiling ($z \geq 1.5\text{m}$) boundary paths:

$$d_{\text{floor}} = \sqrt{x^2 + y^2 + (2h_{\text{head}} + z)^2}, \quad d_{\text{ceiling}} = \sqrt{x^2 + y^2 + (2H_{\text{room}} - 2h_{\text{head}} - z)^2}$$

Boundary early reflections are attenuated by distance divergence and damped with material absorption filters (H_{floor} damped above 3.8 kHz, H_{ceiling} damped above 5.5 kHz).

7. QUANTITATIVE ACOUSTIC METRICS

Configuration	Azimuth θ	Elevation ϕ	ITD (μs)	ILD (dB)	Notch 1 (Hz)	Concha (dB)	Rear Shelf (dB)
Pure Horizontal (Full Left)	−90°	0°	+510.2	−9.3 dB	7,857 Hz	0.0 dB	0.0 dB
Front Center (Eye Level)	0°	0°	0.0	0.0 dB	7,857 Hz	+2.2 dB	0.0 dB
Rear Center (Direct Behind)	180°	0°	0.0	0.0 dB	7,857 Hz	0.0 dB	−4.5 dB
Pure Horizontal (Full Right)	+90°	0°	−510.2	+9.3 dB	7,857 Hz	0.0 dB	0.0 dB
Below Chin (Nadir / Floor)	0°	−40°	0.0	0.0 dB	6,371 Hz	+1.7 dB	0.0 dB
Direct Overhead (Zenith)	0°	+90°	0.0	0.0 dB	11,200 Hz	0.0 dB	0.0 dB
Combined 3D Trajectory	+45°	+45°	−254.8	+6.2 dB	9,528 Hz	+1.6 dB	0.0 dB